

From Theoretical Criteria to Computational Screening: A Framework for Assessing Organic Diradicals as Molecular Qubits

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Abstract

This work develops a computational framework for screening organic molecules as potential molecular spin qubits by mapping the DiVincenzo criteria onto specific quantum chemical calculations. Initial investigation of photogenerated radical pairs in cryptochrome revealed prohibitively large singlet-triplet energy gaps (~ 149 THz), rendering them unsuitable for microwave-based quantum control. This insight directed our focus toward organic diradicals with weak exchange coupling, where $J = 2.81 \text{ cm}^{-1}$ corresponds to microwave-frequency transitions (84 GHz) suitable for qubit manipulation. We implemented a Python-based computational pipeline integrating density functional theory (DFT) derived parameters with quantum dynamics simulations using QuTiP. This framework enables prediction of coherence properties and quantum gate performance, providing a bridge between electronic structure calculations and quantum device metrics. While accurate first-principles prediction of spin relaxation times remains challenging, this work establishes a foundation for high-throughput computational screening of molecular qubit candidates.

1 Introduction

The pursuit of practical quantum computing has motivated exploration of diverse qubit platforms beyond dominant superconducting architectures. While superconducting qubits currently lead in scalability, their requirement for cryogenic operation motivates research into alternative systems including spin qubits, trapped ions, and nitrogen-vacancy centers. Molecular spin qubits represent a particularly promising direction, offering chemical tunability, inherent reproducibility, and potential for self-assembly into scalable arrays.

[1] [2]

Molecular spin qubits leverage unpaired electrons within stable organic frameworks as quantum information carriers. The primary design challenge involves engineering systems with appropriate energy level structures—specifically, achieving singlet-triplet gaps ($\Delta E_{ST} = 2J$) within the microwave frequency regime (1–100 GHz) to enable practical qubit control. Organic diradicals, featuring two weakly coupled unpaired electrons through conjugated bridges, provide an ideal platform for exploring this parameter space.

This work establishes a computational screening protocol grounded in the DiVincenzo criteria for quantum computation [4] [3]. We translate these abstract requirements into specific quantum chemical calculations, creating a pipeline that connects molecular structure to quantum performance metrics. Our approach progresses from initial failed candidates with impractical energy scales to promising systems exhibiting the necessary characteristics for viable molecular qubits. [6]

2 Theoretical Framework: DiVincenzo Criteria

The DiVincenzo criteria provide fundamental requirements for any physical system to function as a quantum computer:

1. A scalable physical system with well-characterised qubits
2. Ability to initialise qubits to a fiducial state ($|0 \cdots 0\rangle$)
3. Qubit-specific measurement capability
4. Long decoherence times relative to gate operation times
5. A universal set of quantum gates

3 Computational Implementation

3.1 Criterion 1: Well-Characterised Qubits

For molecular spin qubits, we identify two quantum states serving as computational basis states $|0\rangle$ and $|1\rangle$. The energy separation must satisfy competing requirements:

- Sufficiently large to prevent thermal population ($\Delta E \gg k_B T$)
- Appropriately small for microwave addressability (1–100 GHz)
- Robust against environmental perturbations
- Singlet-Triplet Gap (ΔE_{ST}): Energy separation between the $S = 0$ singlet and $S = 1$ triplet states, given by $\Delta E_{ST} = 2J$. This defines the qubit operating frequency and must be in the microwave regime (1–100 GHz).

Case Study Application:

- **Cryptochrome:** $J = -4983.99 \text{ cm}^{-1}$ yields $\nu = 149 \text{ THz}$, failing the addressability requirement
- **Diradical 1(c):** $J = 2.81 \text{ cm}^{-1}$ yields $\nu = 84 \text{ GHz}$, satisfying all criteria

[code](#)

3.2 Criterion 2: Initialisation

For photogenerated systems, initialisation occurs naturally through photoexcitation. Computational verification involves:

- TD-DFT identification of accessible excited states
- Confirmation of charge-transfer character
- Verification of spin-correlated radical pair formation

For stable diradicals, we confirm the ground state can be reliably prepared as $|0\rangle$.

3.3 Criterion 3: Qubit-Specific Measurement

We can evaluate three different readout modalities computationally:

1. **Optical:** Transition energies and oscillator strengths via TD-DFT/CASSCF. (This is what we have tried to do by calculating the GHz frequency)
2. **Magnetic:** g -tensors and hyperfine couplings for EPR distinguishability
3. **Chemical:** Spin-dependent potential energy surfaces for reaction yield readout. [\[14\]](#)

3.4 Criterion 4: Long Decoherence Times

While full T_1/T_2 calculations require advanced spin-phonon coupling methods, we employ practical proxies:

- Hyperfine coupling strengths to estimate dephasing rates
- Spin density distributions to assess environmental sensitivity
- For promising candidates, detailed spin-phonon coupling calculations [\[15\]](#), [\[16\]](#), [\[12\]](#), [\[13\]](#)

3.5 Criterion 5: Universal Quantum Gates

Gate operation feasibility requires:

- Microwave-regime transition frequencies
- Substantial transition matrix elements for Rabi driving
- Coherent dynamics demonstrable via master equation simulations
- For two-qubit operations, sufficient exchange or dipolar coupling

4 Case Studies

4.1 Cryptochrome Radical Pair: A Failed Candidate

DFT calculations revealed an exchange coupling $J = -4983.99 \text{ cm}^{-1}$, corresponding to a singlet-triplet gap of $\sim 149 \text{ THz}$.

Assessment:

- **Criterion 1:** Well-defined states exist
- **Criterion 4/5: FAIL** - THz-scale transitions preclude microwave control

The excessive exchange coupling, resulting from strong orbital overlap in the contact radical pair, prevents practical quantum control despite well-defined spin states. cite-crypto, mani

4.2 Organic Diradical 1(c): A Promising Candidate

DFT-derived exchange coupling $J = 2.81 \text{ cm}^{-1}$ corresponds to $\Delta E_{ST} \approx 84 \text{ GHz}$.

Assessment:

- **Criterion 1: PASS** - Pure doublet ground state with appropriate gap
- **Criterion 4/5: PASS** - Microwave-addressable transitions enable gate operations

The designed spatial separation in diradical 1(c) yields an exchange coupling ideally suited for quantum control while maintaining thermal stability.

5 Quantum Dynamics Framework

We developed a quantum dynamics simulator using QuTiP that accepts DFT-derived parameters (J , ZFS, hyperfine couplings) to model qubit performance. The time evolution follows the Lindblad master equation:

$$\dot{\rho} = -\frac{i}{\hbar}[H_{\text{total}}, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (1)$$

where $H_{\text{total}} = H_{\text{exchange}} + H_{\text{ZFS}} + H_{\text{hyperfine}} + H_{\text{Zeeman}} + H_{\text{drive}}$ encompasses all relevant spin interactions.

This framework enables:

- Simulation of Rabi oscillations and gate fidelity
- Magnetic field response analysis for addressability
- Estimation of coherence limits from hyperfine interactions
- Direct connection between molecular structure and quantum performance

[code](#)

6 Conclusion

We have established a computationally rigorous framework for assessing molecular spin qubits that successfully bridges quantum chemical calculations with quantum information science requirements. The pipeline correctly identified why cryptochrome radical pairs fail as qubits while demonstrating the promise of designed organic diradicals.

The key insight—that exchange coupling must yield microwave-frequency transitions while maintaining thermal isolation—provides a clear design principle for molecular qubit synthesis. Our quantum dynamics simulations further validate the approach, demonstrating how DFT-derived parameters translate to measurable quantum performance.

Future work will focus on extending the framework to include more accurate decoherence time predictions and developing machine learning models for rapid property estimation. This computational screening approach represents a crucial step toward the rational design of molecular systems for quantum information processing.

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